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Vellore Institute of Technology
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School of Computer Science Engineering and Information Systems
B.Tech Information Technology
BCSE355L : Cloud Architecture Design
FALL 2026-2027
Course Project -Review II
(Review Date : 07.09.2026 – 11.09.2026)
Link to Upload: https://forms.gle/cyQMu6huDLiHFaZV8

TEAM NAME : Nebula Knight

Team Member(s) with Reg # and Name : Moorthy Chetan; 24BIT0246; Mobile: 9441522812; moorthychetan06@gmail.com

Project Title : CloudPulse: Autonomous Cloud Reliability and Self-Healing Simulator Using AWS Serverless Architecture

1. Introduction

1.1 Background
Cloud computing has become a fundamental platform for modern software applications, where availability, reliability and rapid fault recovery are critical requirements. Conventional monitoring systems can detect failures, but detection alone often requires human intervention, increasing recovery time and operational overhead. CloudPulse demonstrates an event-driven, serverless approach to cloud reliability and self-healing using controlled virtual resources and real AWS control-plane services.

1.2 Problem Statement
Traditional cloud monitoring solutions primarily focus on detecting failures and notifying administrators. A student-scale system rarely demonstrates the complete cycle from controlled failure generation to automated detection, recovery, incident auditing and notification. CloudPulse addresses how a cloud system can automatically detect simulated failures, execute an appropriate recovery strategy, record the incident lifecycle and notify users using a scalable event-driven serverless architecture, without modifying real production infrastructure.

1.3 Literature Survey
The Review-II requirement specifies 10 reputed TR/IF journal sources per student. For this single-member submission, 10 journal sources are included:
1. Scheuner, J. & Leitner, P., “Function-as-a-Service Performance Evaluation: A Multivocal Literature Review,” Journal of Systems and Software, 2020.
2. Cito et al., “A Mixed-Method Empirical Study of Function-as-a-Service Software Development in Industrial Practice,” Journal of Systems and Software, 2019.
3. Yussupov et al., “FaaSten Your Decisions: A Classification Framework and Technology Review of Function-as-a-Service Platforms,” Journal of Systems and Software, 2021.
4. Mampage, A., Karunasekera, S. & Buyya, R., “A Holistic View on Resource Management in Serverless Computing Environments: Taxonomy and Future Directions,” ACM Computing Surveys, 2022.
5. Cassel, G.A.S. et al., “Serverless Computing for Internet of Things: A Systematic Literature Review,” Future Generation Computer Systems, 2022.
6. “Real-time Resource Scaling Platform for Big Data Workloads on Serverless Environments,” Future Generation Computer Systems, 2020.
7. “Serverless Execution of Scientific Workflows: Experiments with HyperFlow, AWS Lambda and Google Cloud Functions,” Future Generation Computer Systems, 2020.
8. “AFCL: An Abstract Function Choreography Language for Serverless Workflow Specification,” Future Generation Computer Systems, 2021.
9. Brogi, A. et al., “Self-Healing Trans-Cloud Applications,” Computing, 2022.
10. “Enhancing the Analysis of Software Failures in Cloud Computing Systems with Deep Learning,” Journal of Systems and Software, 2021.

2. Overview and Planning

2.1 Proposed System Overview
CloudPulse is a serverless cloud reliability and self-healing simulator in which virtual resources generate controlled failures; CloudWatch detects degraded metrics, EventBridge routes events, Lambda performs recovery, DynamoDB records incidents, SNS sends notifications, and a React dashboard presents live state. The core flow is Failure Simulation → CloudWatch Metric/Alarm → EventBridge Rule → Recovery Lambda → DynamoDB Incident/Audit → SNS Notification → Dashboard Update.

2.2 Challenges

Reliable event correlation; idempotent recovery; deterministic failure scenarios; safe AWS usage; observability; CORS/security controls; asynchronous events; and cost-conscious serverless deployment.

2.3 Assumptions
Failures are simulated logically; no real EC2/RDS infrastructure is modified. AWS control-plane services are real. The demonstration targets a small educational workload. Failure scenarios are deterministic and recovery strategies are predefined.

2.4 Architecture Specifications
Frontend: React + TypeScript. API/Application: API Gateway HTTP API + FastAPI on AWS Lambda via Mangum + boto3. Monitoring: CloudWatch custom metrics and alarms. Event processing: EventBridge. Recovery: Lambda with deterministic strategies and bounded attempts. Data: DynamoDB. Notifications: SNS. Failure handling: SQS DLQ. Deployment/hosting: AWS SAM, CloudFormation and S3. IAM provides access control.

2.5 Hardware Requirements
Laptop/desktop with a modern web browser, minimum 4 GB RAM recommended, stable internet connection, keyboard and mouse. Local machine is used for development, testing, AWS CLI/SAM and dashboard access; core application infrastructure runs on AWS.

2.6 Software Requirements
AWS Lambda, API Gateway, DynamoDB, CloudWatch, EventBridge, SNS, SQS, S3, IAM, CloudFormation; Python 3.12; FastAPI; boto3; Mangum; React; TypeScript; Node.js 20; AWS SAM; Git/GitHub.

3. System Design

3.1 High-Level Design
React Dashboard → API Gateway HTTP API → FastAPI Lambda → DynamoDB. A simulator periodically publishes custom CloudWatch metrics. CloudWatch Alarms → EventBridge → Recovery Lambda → DynamoDB/SNS. The system is event-driven and serverless.

3.2 Low-Level Design
Virtual resources: VM-001, API-001, STORAGE-001, DB-001. Five deterministic scenarios are implemented:
FS-01 High CPU Utilization → SCALE_OUT
FS-02 Service Failure → SERVICE_RESTART
FS-03 Storage Exhaustion → STORAGE_CLEANUP
FS-04 Network Latency → NETWORK_REROUTE
FS-05 Service Downtime → FAILOVER
Recovery uses a dispatcher, conditional DynamoDB writes for idempotency and bounded recovery attempts. The lifecycle is HEALTHY → WARNING → FAILURE_DETECTED → RECOVERY_INITIATED → RECOVERY_IN_PROGRESS → RECOVERED, with RECOVERY_FAILED → MANUAL_INTERVENTION_REQUIRED as the failure path.

4. Implementation : 80%

Approximately 80% of the planned system has been implemented and deployed on AWS. The serverless infrastructure has been created using AWS SAM/CloudFormation, including Lambda functions, API Gateway HTTP API, DynamoDB tables, CloudWatch metrics and alarms, EventBridge rules, SNS, SQS DLQ, S3 and IAM roles.

The FastAPI backend is integrated with Lambda through Mangum, with APIs for health, resources, incidents and metrics. The failure simulation engine supports five deterministic scenarios and maps them to automated recovery strategies. The self-healing engine performs event-driven recovery with state transitions, conditional writes and bounded attempts. The React/TypeScript dashboard provides resource health, failure simulation, active incidents, incident details, recovery activity, reliability metrics and architecture visualization. Automated testing, TypeScript validation, SAM validation and security checks have been completed during implementation. The working flow is: Simulated Fault → Detection → Autonomous Recovery → Incident Persistence → Notification → Dashboard Update.

Additional Related Information

Innovation: CloudPulse combines a safe educational digital-twin concept with real AWS event-driven services to demonstrate autonomous cloud reliability without persistent EC2/RDS infrastructure.

Reliability metrics: Incident Count, Recovery Success/Failure Rate, Average Recovery Time/MTTR, Detection Time, MTBF and Health Distribution are derived from incident/resource records.

Safety boundary: failures are logically simulated; real AWS monitoring, event routing, Lambda execution, DynamoDB persistence, SNS notification, SQS DLQ, API Gateway and S3 hosting are used for the control plane.

Current implementation note: the deployed system is intended as an academic MVP; production-scale distributed tracing, very large-fleet indexing and true wall-clock recovery measurement remain outside the current scope.

For Evaluation only
Status : ACCEPTED / REJECTED

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Project Title:	**CloudPulse: Autonomous Cloud Reliability and Self-Healing Simulator Using AWS Serverless Architecture**
Team Name	**Nebula Knight**

Project Team

S.No	Register Number	Student Name	Signature	Guided By
1	24BIT0246	Moorthy Chetan		

Team Member(s) Contribution and Performance Assessment

Components	Student 1	Student 2	Student 3
Literature Survey (03)			
Design (06)			
Implementation (06)			
Total (15)			
Components			

Expectation for Next Reviews	Comments

Name & Signature of the Evaluator